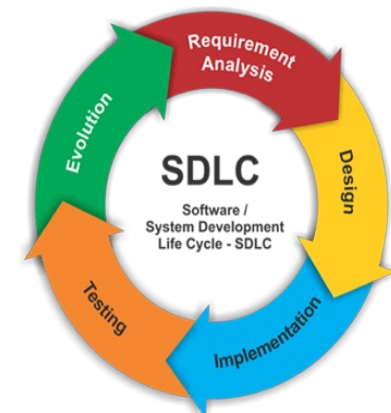


Chapter 1: Software Development Life Cycle - 1



What is Software Development Life Cycle (SDLC)?



What is Software Development Life Cycle (SDLC)?

- Software Development Life Cycle is a process used by system developers to design, develop and test software.
- SDLC consists of a detailed plan describing how to develop and maintain a particular software.
- It aims to produce a high quality software that meets or exceeds customer expectations, completion within deadline and costs.

Typical Phases in SDLC



SDLC Phases:

Planning and Requirement Analysis

- Identify and capture stakeholder requirements (*business requirements*)
 - surveys
 - customer interviews
 - etc.
- Build multiple use cases to describe each action that a user will do in the new system.
- Functional Requirements: Specifies what the system should do
 - E.g. System must display the result of two numbers' addition
- Non-functional requirements: Describes how the system should do
 - E.g. System has to display the result within 5 seconds



SDLC Phases:

Design the Program / System Design

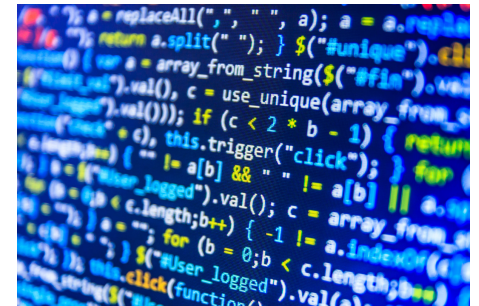
- Define solution system, based on requirement and analysis decision.
- Build Prototypes to show the client what the end product will look like.

SDLC Phases:

Coding the Program / Implementation

Where actual development of software starts

- Developers code the software based on the Requirements Documents developed during the User Requirement gathering phase
- Programming language used with respect to the type of software being developed
 - E.g. Data Science Model – Python
 - E.g. Android apps - Java



SDLC Phases:

Testing the Program



Once the coding has been completed, the software will be migrated to a test environment for testing

Different types of testing will be performed:

- Unit Testing - Individual units/components of a software are tested
- Integration Testing - Individual components are combined and tested as a group
- System Testing - A complete and integrated software is tested as a whole
- Acceptance Testing - Formal testing with respect to user needs and requirements

Black Box & White Box Software Testing Methods

BLACK BOX TESTING

- The internal structure/ design/ implementation of the item being tested is **NOT known to the tester**
- Done by software testers
- No knowledge or implementation is needed
- Outer or external software testing

WHITE BOX TESTING

- The internal structure/ design/ implementation of the item being tested is **known to the tester**
- Done by software developers
- Knowledge of implementation is required
- Inner or internal software testing

SDLC Phases:

Deploy and Maintain the Program

- Completed product will be released formally as a whole or in stages according to customer requirement.
- Product may also be released to a limited segment and tested in the real business environment for user acceptance testing.
- User feedback or any problems surfaced will be solved and maintained by the developer company.

Activity – Research about “Regression Testing”

Is **regression testing** part of UAT?

Students to do a research and share answers during lesson.

Examples of SDLC Models

- Waterfall Model
- Iterative Model
- Spiral Model (Meta Model)
- Prototype Model
- V-Model
- RAD (Rapid Application Development) Model
- Agile Model
- DevOps Model



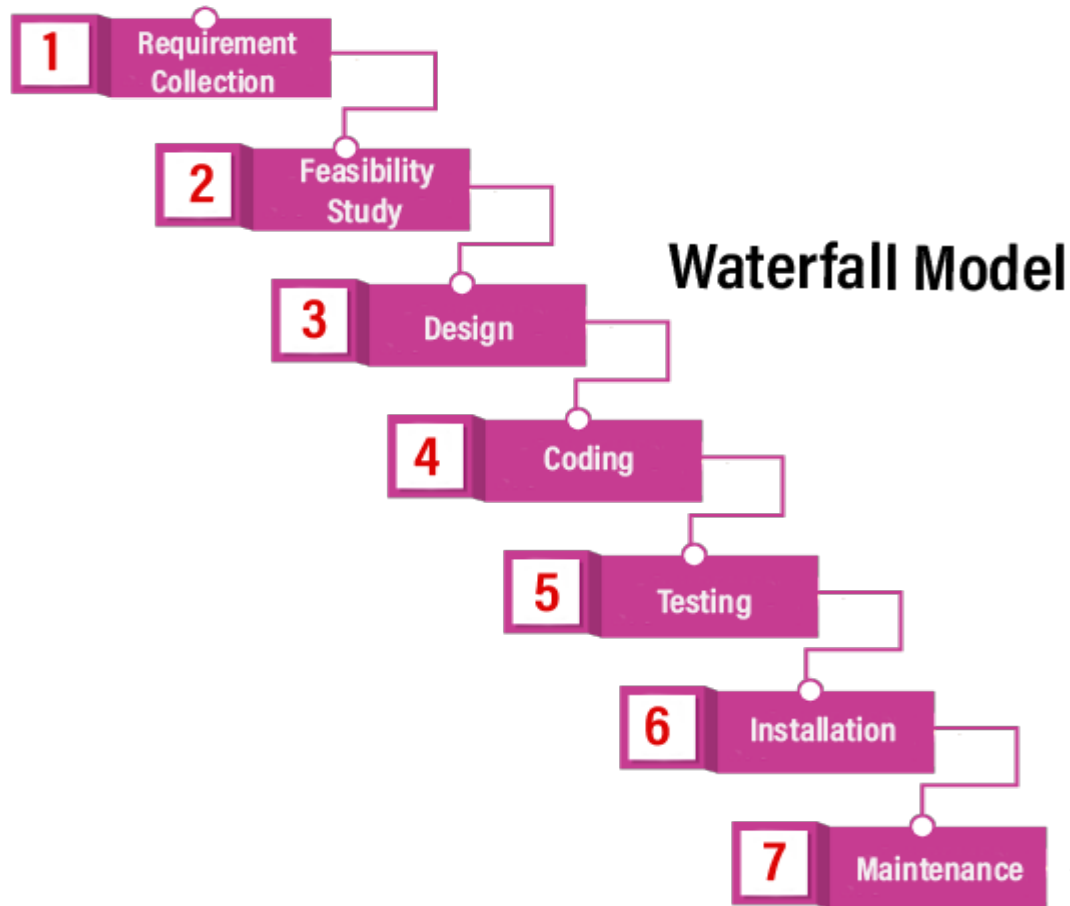
Waterfall Model

Waterfall approach was first SDLC Model to be used widely in Software Engineering to ensure success of the project.

In Waterfall Model, each phase must be completed before the next phase can begin and there is no overlapping in the phases.



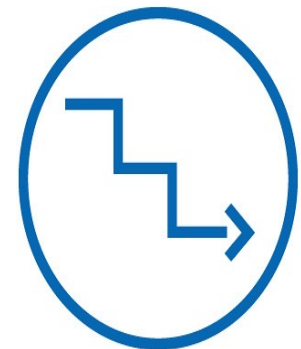
Waterfall Model



Waterfall Model

Waterfall SDLC Method includes:

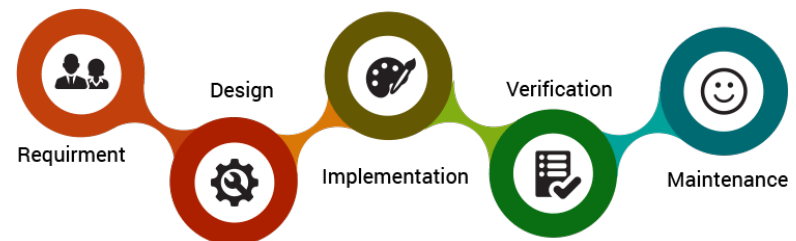
- Requirement Collection & Analysis
- Feasibility study
- Design / System Design
- Implementation / Coding
- Integration and Testing
- Deployment of system / Installation
- Maintenance



Waterfall Model

Advantages

- Simple and easy to understand and use.
- Easy to manage due to the rigidity of the model. Each phase has specific deliverables and a review process.
- Phases are processed and completed one at a time.
- Works well for smaller projects where requirements are very well understood.
- Clearly defined stages.



Waterfall Model

Disadvantages

- No working software is produced until late during the life cycle.
- High amounts of risk and uncertainty.
- Not a good model for complex and object-oriented projects.
- Poor model for long and ongoing projects.
- Difficult to measure progress within stages.



Building a prototype in waterfall model is feasible or not? Give an example.



Summary

1. **SDLC** - It is a process used by system developers to design, develop and test software.
2. **SDLC** aims **to produce a high quality software** that meets or exceeds customer expectations, completion within deadline and costs.
3. **Planning & Requirement Analysis** – Identify and capture stakeholder requirements (*business requirements*) using customer interviews and surveys.
4. **Functional Requirements**: This specifies something the system should do.
5. **Non-functional requirements**: This describes how the system should work/behave.

Summary

7. **Coding Phase** - Developer codes the software based on the Requirements Documents developed during the User Requirement gathering phase.

8. **Unit Testing** - Individual units/components of a software are tested.

9. **Integration Test** - Individual components are combined and tested as a group.

10. **System Testing** - A complete and integrated software is tested as a whole.

11. **Acceptance Testing** - Formal testing with respect to user needs and requirements.

Summary

12. **Black Box Testing** is a software **testing** method in which the internal structure/ design/ implementation of the item being tested is **NOT known to the tester**.

13. **White Box Testing** is a software **testing** method in which the internal structure/ design/ implementation of the item being tested is **known to the tester**.

14. **Examples of SDLC models** - Waterfall Model, Iterative Model, Spiral Model (Meta Model), Prototype Model, V-Model, RAD (Rapid Application Development) Model, Agile Model, DevOps Model.

15. Focus on **advantages & disadvantages of Waterfall Model**.

Reference

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